

Vanessa Ching

vanessazqching@gmail.com | (678) 779-0547 | [linkedin.com/in/vanessa-ching](https://www.linkedin.com/in/vanessa-ching) | vanessaching.github.io/personal-portfolio

EDUCATION

Johns Hopkins University, Whiting School of Engineering

B.S. in Computer Science, Minor in Film and Media Studies — May 2028

Baltimore, MD

Relevant Coursework: Data Structures, Computer Systems Fundamentals, Intermediate Programming, Gateway Computing: Python, Mathematical Foundations for Computer Science

EXPERIENCE

Undergraduate Researcher

Jan 2026 – Present

Sports Analytics Research Group, JHU

Baltimore, MD

- Built Voice Scorekeeper, a real-time AI baseball scorekeeping platform (React, TypeScript, Node.js, Tailwind CSS) commissioned for the Atlantic League of Professional Baseball, with a league presentation planned
- Owned the backend transcription/LLM pipeline: tuned Whisper vocabulary handling to fix baseball-specific transcription errors and benchmarked LLM backend, reducing inference/transcription latency from ~21s to ~7s via prompt reduction and local model support
- Validated structured event output against MLB Gameday play-by-play feeds and archival radio broadcasts, matching manual scorekeeping accuracy at significantly faster turnaround

Research Intern

May 2026 – Aug 2026

Georgia Tech Research Institute (GTRI)

Atlanta, GA

- Integrated Unreal Engine and computer vision pipelines (OpenCV, YOLO, RT-DETR) into synthetic simulation environments with KLV/UAS metadata support for autonomous perception testing
- Developed a multi-drone vehicle tracking system evaluating AI perception across terrain, lighting, and occlusion conditions, using UnrealCV for real-time frame streaming and a natural-language object-query system for open-vocabulary target search

Research Assistant

Aug 2025 – Jan 2026

Social Cognitive AI Lab, JHU

Baltimore, MD

- Developed procedural urban simulation environments in Unreal Engine 5 and Python for embodied AI navigation experiments
- Designed dynamic multimodal scenarios with environmental variation, moving obstacles, and spatial audio cues to support agent reasoning research

Software Engineering Intern

May 2025 – Aug 2025

Frenalytics

Remote

- Built an AI-powered WordPress reading assistant platform supporting 1,500+ students with literacy challenges
- Designed PHP REST APIs and MySQL-backed session infrastructure for avatar customization, progress tracking, and analytics

PROJECTS

F1 Tire Degradation & Pit Stop Strategy Simulator

Python, FastAPI, React, Vite, Tailwind, FastF1

- Built a full-stack race strategy simulator pulling real historical race data via FastF1, modeling per-compound tire degradation curves with temperature coefficients fit from real per-lap track temperature data rather than assumed constants
- Implemented multi-stop undercut/overcut strategy simulation and pit-loss estimation split by green-flag vs. safety-car conditions

AI Readezzy Reading Assistant

PHP, MySQL, JavaScript, WordPress

- Built an AI-assisted reading platform featuring dynamic avatar systems, session persistence, REST APIs, and relational database-backed personalization workflows

ACTIVITIES & LEADERSHIP

Pi Beta Phi Fraternity for Women — VP of Operations

Sep 2025 – Present

- Coordinate chapter operations and cross-functional officer initiatives for a 60+ member organization

Hopkins Student Organization for Programming (HOP) — Fine Arts Co-Chair

Sep 2024 – Present

- Co-lead logistics and vendor coordination for campus-wide fine arts programming serving 5,600+ students

Matriculate — Advising Fellow

Jan 2026 – Present

- Mentor high-achieving, low-income high school students through the college application process

Student Government Association — Class Council Representative

Jan 2026 – Present

- Execute large-scale class initiatives for a cohort of 1,000+ students

SKILLS

Languages: Python, JavaScript/TypeScript, C++, C, Java, SQL, PHP

Frameworks & Tools: React, Node.js, Express, FastAPI, Tailwind CSS, REST APIs, MySQL, Supabase, TCP sockets, Git, Linux

AI & Simulation: LLM APIs (OpenAI/Ollama/Groq), Whisper, OpenCV, YOLO, Unreal Engine 5